

Polynomial Interpolation (piecewise)

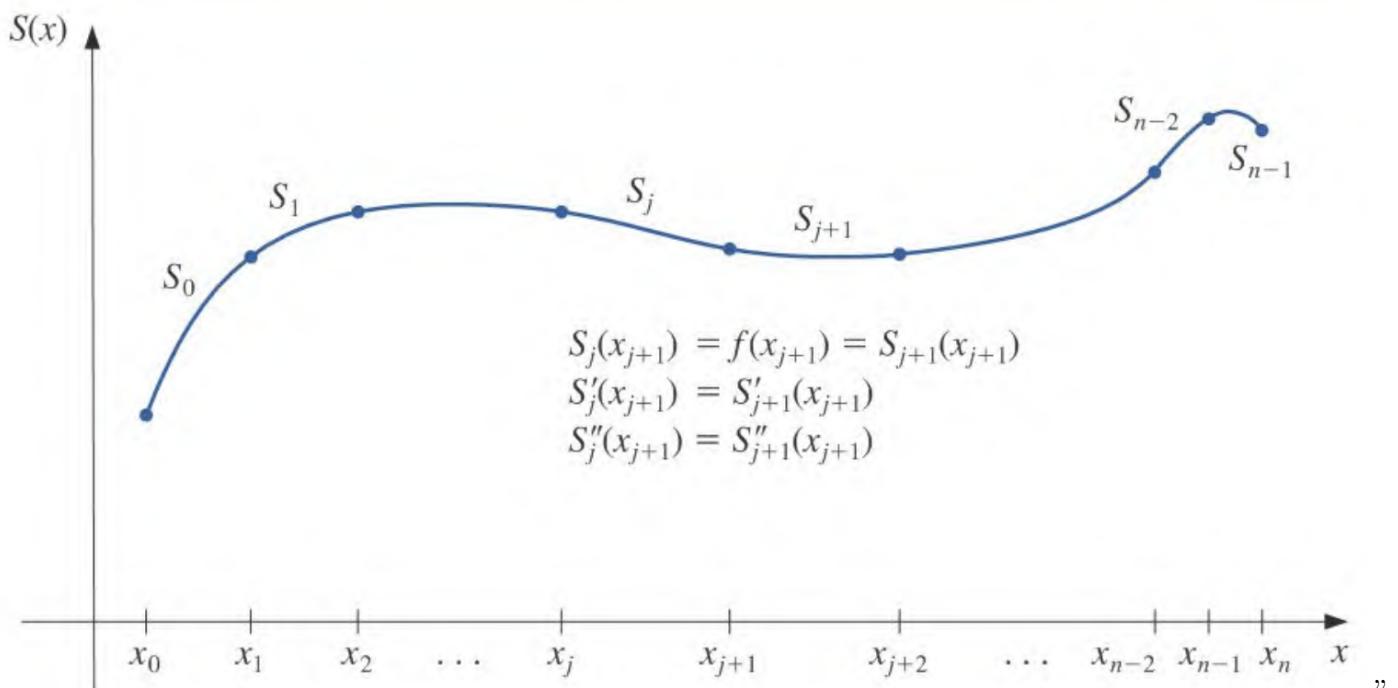
1.3 Cubic Splines

Interpolation relying on a single high-degree polynomial—such as in Lagrange interpolation—can lead to erratic oscillations, especially when the number of points exceeds a dozen. This phenomenon highlights why very high-degree polynomials should generally be avoided in favor of piecewise or spline interpolation for better stability and accuracy.

In piecewise interpolation, we divide the interval into small subintervals and construct a low degree (0 to 3) interpolation polynomial on each subinterval, which is called a **spline**.

1. **Piecewise Linear Interpolation:** Uses straight lines to connect points but is not differentiable at the interpolating points.
2. **Piecewise Quadratic Interpolation:** Uses quadratic functions on each subinterval. However, enforcing derivative conditions at boundary points x_0 and x_n can be problematic.
3. **Cubic Spline:** A degree-3 spline that is twice differentiable at interpolation points, with well-defined boundary conditions. It is the most widely used piecewise polynomial approximation due to its smoothness and stability.

A simple Definition of Cubic Spline Interpolation: Given a set of data points, a cubic spline interpolant is a piecewise cubic function that passes through the given points while maintaining continuous first and second derivatives (C^2 continuity) across the entire domain.



Detailed Definition: Cubic Splines

Given as set of data (x_i, y_i) , with $a = x_0 < x_1 < \cdots < x_n = b$,

- (a) $S(x)$ is a cubic polynomial, denoted by $S_j(x)$, on the subinterval $[x_j, x_{j+1}]$ for each $j = 0, \cdots, n-1$.
- (b) $S_j(x_j) = f(x_j)$ and $S_j(x_{j+1}) = f(x_{j+1})$, $j = 0, \cdots, n-1$.
- (c) $S_j(x_{j+1}) = S_{j+1}(x_{j+1})$, $j = 0, \cdots, n-2$.
- (d) $S'_j(x_{j+1}) = S'_{j+1}(x_{j+1})$, $j = 0, \cdots, n-2$.
- (e) $S''_j(x_{j+1}) = S''_{j+1}(x_{j+1})$, $j = 0, \cdots, n-2$.
- (f) One of the following sets of boundary conditions (BCs) is imposed:
 - (i) $S''(x_0) = S''(x_n) = 0$, **natural or free BC**
 - (ii) $S'(x_0)$ and $S'(x_n)$ are equal to some values; **clamped BC**.

When the natural BC is used, the result is **natural cubic spline**, and its graph approximates the shape, which a long flexible rod would assume if forced to go through each of the data points. Similarly, when the clamped boundary conditions occur, the spline is called a **clamped cubic spline**. In general, clamped boundary conditions lead to more accurate approximations since they include more information about the function.

To construct the cubic line interpolant for a given function f , the conditions in the above definition are applied to the cubic polynomials. There is a unique natural cubic spline for any data set. That is, the cubic spline S interpolating (x_i, y_i) with $a = x_0 < \cdots < x_n = b$ and satisfying $S''(a) = S''(b) = 0$.

Example 1. Construct a natural cubic spline passing through $(1, 2)$, $(2, 3)$, and $(3, 5)$.